

20. Light of wavelength 720 nm from a laser X is incident normally on a diffraction grating and a diffraction pattern is observed. Light from a laser Y is then also incident normally on the same grating. The third-order maximum due to laser Y is seen at the same place as the second-order maximum due to laser X.

What is the wavelength of the light from laser Y?

- A** 480 nm **B** 540 nm **C** 720 nm **D** 1080 nm

Answer: A

$$d \sin \theta = n \lambda$$

For same grating and at the same position, $d \sin \theta = \text{constant}$

$$\text{So, } 2\lambda_X = 3\lambda_Y \quad \lambda_Y = \frac{2}{3} \times 720 = 480 \text{ nm}$$